

The cost of a nutritious diet in Afghanistan, Bangladesh, Pakistan, and Sri Lanka[☆]

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World Bank, United States

ARTICLE INFO

Keywords:
Food prices
Diets
Nutrition
Dietary recommendations

JEL classification:

Q1
I1
I3

ABSTRACT

The high cost of a nutritious diet can worsen nutrition outcomes especially among low-income households. Yet little is known about the cost of a nutritious diet in South Asia, where malnutrition in multiple forms remains high. We use data from Sri Lanka, Pakistan, Afghanistan, and Bangladesh to calculate the minimum cost of meeting recommended food-based dietary guidelines, or the Cost of a Recommended Diet (CoRD). We further compare CoRD to actual food expenditures. Across these countries, we find that majority of households spend less on food than the CoRD estimate, that an average household overspends on staples and protein foods and underspends on dairy and vegetables, and that subnational variation in CoRD is to an extent driven by the higher variability in the price of vegetables relative to that of starchy staples. Our results suggest a need to shift food policy focus away from energy-dense towards nutrient-rich foods.

1. Introduction

Around the world, the cost of nutritious foods, such as fruits, vegetables, and animal-source foods, is typically higher than the cost of more calorie-dense micronutrient-poor foods, such as starchy staples, oil and sugar. In some parts of the world, this gap in price has been increasing over time (Bouis, 2017). Moreover, the cost of nutritious foods tends to be more variable across geographic locations. Nutritious foods are often highly perishable and less tradable. As such, their prices are largely determined by local productivity and value chain efficiency (Headey, 2017; Monsivais et al., 2010).

The relatively high cost of nutritious foods can affect nutrition outcomes. While price elasticities differ across countries and foods, higher relative prices can generally result in reduced consumption of nutritious foods (Miller et al., 2016; Andreyeva et al., 2010; Green

et al., 2013). The cost of nutritious diets has also been linked to the quality of overall diets and to nutrition outcomes, such as stunting and obesity (Beydoun et al., 2011; Grossman et al., 2013; Headey, 2017). Importantly, the poor are more likely affected by the higher cost of nutritious foods. They are more sensitive to prices, and as such are likely to choose cheaper, energy-dense diets (Miller et al., 2016; Green et al., 2013; Darmon and Drewnowski, 2015). While the cost of healthy and unhealthy foods is not the only factor determining diets and nutrition outcomes, it is an important one and likely more binding for the poor than the non-poor.

Our study focuses on describing the cost of recommended nutritious diets in a few countries in South Asia, a region highly affected by malnutrition in all its forms.¹ The region has among the highest burdens of child undernutrition in the world, with 36 percent stunted and 16 percent wasted (UNICEF, WHO, and World Bank, 2017).² Stunting rates

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¹ Looking at micronutrient deficiencies in South Asia, 42 percent of children under age five are vitamin A deficient and 49 percent of women are anemic; the figures are greater in India and Pakistan. Looking at overconsumption and diet-related non-communicable disease, 24 percent of women are overweight, more than 25 percent of adults have raised blood pressure, and the region has one of the highest rates of raised blood glucose (9–10 percent) (Development Initiatives, 2017).

² Stunting is a measure of low height for age, an indicator of chronic undernutrition; wasting is a measure of low weight for height, an indicator of acute malnutrition.

are not only high, but they are also highest among the poor. While there are multiple causes of malnutrition, the coexistence of undernutrition, micronutrient deficiencies, and markers of diet-related disease signal that poor diets are widespread. In South Asia, 13% of people experience severe food insecurity (FAO, 2017). Many more lack access to adequate diets. Dietary data available for Afghanistan show that dietary diversity is very low; only 8 percent of women achieve the *minimum* dietary diversity,³ meaning that very few women are likely meeting nutritional needs (WFP, 2017a). Only 22% of young children age 6–23 months receive minimum dietary diversity (at least 4 of 7 food groups) in India and Pakistan (Development Initiatives, 2018). Cost likely plays a significant role in access to nutritious diets. Other work has found that across 15 countries in Asia and the Pacific, the price of food has increased faster than overall inflation, and specifically the price of fruits and vegetables has increased faster than the rest of the food category (Dawe and Lee, 2017).

We calculate the cost of meeting dietary recommendations in four countries in the region—Sri Lanka, Pakistan, Afghanistan, and Bangladesh. The Cost of a Recommended Diet (CoRD) is a metric of the minimum cost of meeting food-based dietary guidelines (FBDGs) and uses data on food prices and national FBDGs (Cost of Nutritious Diets Consortium 2018). FBDGs are a representation of diets that include but also go beyond basic nutrients to encompass foods toward the aim of overall protection of health. Moreover, the use of FBDGs can improve policy coherence, by showing whether such government recommendations for dietary intake based on expert opinion and in-country dialogue are affordable for consumers. We further compare CoRD against actual food expenses in each of these countries, for the total diet and its component food groups, and analyze subnational variation. To aid policymakers in gauging where cost of nutritious diets is a much more paramount constraint, a key descriptive contribution of this paper is to map subnational differences in the proportion and number of households who are unlikely to be meeting the recommended diet due to its high cost.

This paper complements emerging work on the cost of nutritious diets. One method calculates an index of the cost of minimum dietary diversity, following the minimum dietary diversity score for women (MDD-W) (Masters et al., 2018). Several others calculate the cost of achieving nutrient adequacy (Deptford et al., 2017; Cofer et al., 1962; Masters et al., 2018). A prominent methodology among the latter is the Cost of the Diet (CoD), developed by Save the Children UK, and implemented in Fill the Nutrient Gap (FNG) analyses done by the World Food Programme (WFP, 2017b). CoD is a linear programming tool which uses food price data, and models the cost of a theoretical, simulated diet (food basket) that satisfies nutrient requirements of a household at the minimal cost, based on the availability, price, nutrient content, and acceptability of local foods (Deptford et al., 2017). CoD has been calculated in various countries in South Asia, and tends to be higher than the cost of an energy only diet; it is also higher in certain seasons and varies across geographic location (Chastre et al., 2007; Pakistan, Ministry of Planning, Development & Reform, and WFP 2016; WFP, MED and HARTI 2014; Biehl et al., 2016; WFP, 2017c; WFP, 2018).

Our CoRD calculation builds on the methodologies described above, with a few important differences. While CoD bases its calculations on meeting essential vitamins and minerals, CoRD reaggregates food price information based on food-based dietary guidelines (FBDGs). Such guidelines are a representation of diets that include but also go beyond basic nutrients to encompass foods toward the aim of overall protection of health. Few other studies have attempted to calculate the cost of dietary recommendations (Mulik and Haynes-Maslow, 2017). The use of FBDGs can improve policy coherence, by showing whether

government recommendations for dietary intake based on expert opinion and in-country dialogue are affordable for consumers. In addition, our methods use simpler calculations in contrast to the more stringent data requirements and linear programming method of the CoD approach. Countries may calculate CoRD from existing food price monitoring data, and can track these easily with little added cost or expertise.

We offer accessible methods for policymakers to track the cost of a nutritious diet by harnessing existing food price data. These methods can be used by policymakers along with the food price metrics they typically collect, such as CPI and the price of basic staple commodities. The additional information serves to broaden awareness and attention to the cost of nutritious diets, and thereby enables better-informed policy decisions.

2. Data and methodology

2.1. Methodology

CoRD is a method to estimate the minimum daily cost of achieving the recommended diet for an average adult, based on FBDGs (Cost of Nutritious Diets Consortium 2018). To calculate CoRD, two pieces of information are needed, namely the price of each food item (discussed in Section 2.2) and information on FBDGs (discussed in section 2.3). We use two sources of food price data: CPI data and national household survey data, discussed below in more detail.

Calculating CoRD involves the following steps. First, each food in a list, is categorized according to the food groups established under the FBDGs. National Statistics Offices are one primary source of food price data, and typically organize their food list by COICOP categories (an international system of Classification of Individual Consumption according to Purpose); Table 1 shows how COICOP categories are converted into FBDG categories. Food items under the categories sweets, spices, and beverages (other than milk) are excluded from the CoRD calculation because they are not required in recommended diets. Second, duplicate foods are eliminated; for example, it is not uncommon for a country to measure several different kinds of the same staple, such as rice (local), rice (imported), rice (local inferior). Only the lowest-cost duplicate item is kept. Third, all food item prices are converted into price per edible serving, which is calculated by multiplying the reported price of a food item by the following price conversion factor:

$$\text{price conversion factor} = \frac{\text{grams per serving}}{\text{reported unit of food item}} / \text{edible portion}$$

where grams per serving is based on the regional FBDG (discussed below in section 2.3) and unit of food item is given in the food price data.

Fourth, for each food group we take the average of the two items with the lowest price per edible serving; except for vegetables, where the three cheapest items are selected, one of which must be a dark green leafy vegetable (DGLV), as specified in the regional FBDGs. We choose more than one low-cost item, because FBDGs are based on the concept of diversity among and within food groups. We choose lowest cost items, as our objective is to calculate the minimum cost of meeting the recommended diet. Other research calculating the cost of meeting dietary guidelines has used average prices of all items in the food group category (Mulik and Haynes-Maslow, 2017). We chose the lowest-cost items to avoid sensitivity to very expensive food items; an alternative is to select the median cost items, which is also presented in the annex as a robustness check. Fifth, the average price per edible serving for each food group is multiplied by the average of the lower- and upper-bound of the number of servings recommended for that food group. This is the cost for meeting the recommendations for each food group. Finally, to calculate CoRD, the costs of all food groups are summed.

³ Equal to five of 10 food groups defined in the minimum dietary diversity for women (MDD-W) indicator.

Table 1
Guide for grouping food and beverage items into FBDG categories.

FBDG categories	COICOP categories included
Starchy staples	01.1.1 - Bread and cereals 01.1.7 - Vegetables: Only starchy root crops such as potatoes, sweet potatoes, yam
Vegetables	01.1.7 - Vegetables: All vegetables except pulses and starchy roots
Fruits	01.1.6 - Fruit: All fruit except nuts and seeds
Protein foods (Pulses/Fish/Meat/Eggs)	01.1.7 - Vegetables: Only pulses 01.1.2 - Meat 01.1.3 - Fish and seafood 01.1.4 - Milk, cheese and eggs: Eggs only
Dairy	01.1.4 - Milk, cheese and eggs: Milk and cheese only
Oils	01.1.5 - Oils and fats (excluding margarines/vanaspati)
Not included in FBDG	01.1.8 - Sugar, jam, honey, chocolate, and confectionery 01.1.9 - Food products n.e.c. (not elsewhere classified) 01.2.1 - Coffee, tea, and cocoa 01.2.2 - Mineral waters, soft drinks, fruit, and vegetable juices

Note: Occasionally an item is misclassified in a country's CPI lists (for example, baby formula or ice cream is classified as dairy); the detailed rules for COICOP classifications are used for categorizing foods into FBDG categories.

3. Data on food prices

3.1. CPI price data

For three of the four countries discuss in this paper, Sri Lanka, Pakistan, and Afghanistan, we use prices for each item in the list of food and nonalcoholic beverages that is collected for one month of the CPI.⁴ CPI data for Bangladesh was not available from the government at the time of the study. The data for Sri Lanka are from December 2017, for Pakistan June 2017, and for Afghanistan September 2017. For consistency, we convert our CoRD calculation to the typical month where food prices are lowest—this is March for Sri Lanka and Afghanistan, and February for Pakistan.⁵ To convert the month of price data available to the typical lowest-cost month, we collected food CPI data for these three countries and regressed food CPI on month of year dummies separately for each country to identify the typical month with the lowest food prices. We then calculate the ratio between the month for which CPI data is available to the identified typical month with the lowest prices. Then we divide CoRD by this ratio to arrive at the adjusted CoRD for the typical month of the lowest prices for each country. Note that the CoRD calculations do not vary much with this CPI adjustment, suggesting that the estimates are fairly stable.

The main motivation for using CPI data is to demonstrate that existing food price monitoring data can be used by policymakers to calculate CoRD. CPI price data are not representative of the prices faced by certain consumers, however, especially those living in remote rural areas. To investigate subnational variation in food prices, we turn to household surveys as our primary source of food price data.

3.2. Household survey price data

There are two main advantages of using price data from household surveys. First, while household surveys are collected less frequently than CPI data, the data collected in household surveys are

representative at a certain subnational level, allowing us to evaluate subnational spatial variation in CoRD. Second, from household surveys, we can also calculate actual household expenditures on food. Comparing CoRD to food expenses is a crude but important initial step at gauging affordability (or constrained affordability), as we can estimate how many households spend less on food than what it would have cost (at the very least) to meet the minimum recommended diet. To make it comparable to CoRD, our measure of food expenses similarly excludes expenses on food items under the categories sweets, spices, and beverages.⁶

Food prices from household surveys are deduced from unit values in the food consumption module. Specifically, to calculate a unit value for a given food item, we divide the total amount spent by the total amount consumed for each household. We then take the median of this unit value across households nationally, as well as across households in a subnational level. In our case, we choose the lowest subnational level at which the household survey is representative. Taking the median food item unit value at a national or subnational level largely addresses the typical concerns with using unit values, such as measurement error and its spurious negative correlation with quantities and that unit values typically reflect the quality of a food item (Deaton, 1988).

In this paper, we calculate and use food price data from four household surveys: Sri Lanka 2016 Household Income and Expenditure Survey (HIES) (with a sample of 21,756 households), Pakistan 2015–16 Household Integrated Income Consumption Survey (HIICS) (with 24,328 households), Afghanistan 2011 Living Conditions Survey (ALCS, otherwise known as the National Risk and Vulnerability Survey (NRVA)) (with 20,828 households), and Bangladesh 2016 Household Income and Expenditure Survey (HIES) (with 45,841 households). All household surveys used here are nationally representative. Fig. 1 shows the geographical coverage of data. A key caveat is that for the Afghanistan 2011 ALCS, unit values were not calculated from the survey. Instead, a separate market price survey was conducted, which is then used alongside the household reported consumption quantities to calculate household food expenditures. The 2011 ALCS collected prices in

⁴ Data for Sri Lanka came from the Department of Census and Statistics, Prices and Wages Division, for Pakistan from the Pakistan Bureau of Statistics, and for Afghanistan from the Central Statistics Organization, National Accounts Department, Price Statistics unit.

⁵ The month with the lowest price for each country is determined using monthly food CPI data for Sri Lanka 60 months from January 2013 to December 2017, for Pakistan 108 months from July 2008 to June 2017, and for Afghanistan 69 months from March 2011 to November 2016. If looking only at 2017, the month with the lowest food prices was March in Sri Lanka (which is the same as the typical lowest food price month), but it was January for Pakistan and Afghanistan.

⁶ In Sri Lanka, we use the per adult male equivalent (AME) spatially adjusted food expenses (at the district level). In Pakistan, we use the per AME spatially adjusted food expenses (at the PSU level). In Afghanistan, we use the per AME spatially adjusted food expenses (at the region level). In Bangladesh, we use per AME spatially adjusted food expenses (at the stratum level). Household weights are used to calculate the proportion of households spending less on food than CoRD. Ideally, country-specific AME scales should be used to calculate per AME food expenses for each country. However, AME scales for only Bangladesh and Pakistan were available (Waid et al., 2017). Therefore, AME scales for Bangladesh was applied to Sri Lanka and Afghanistan.

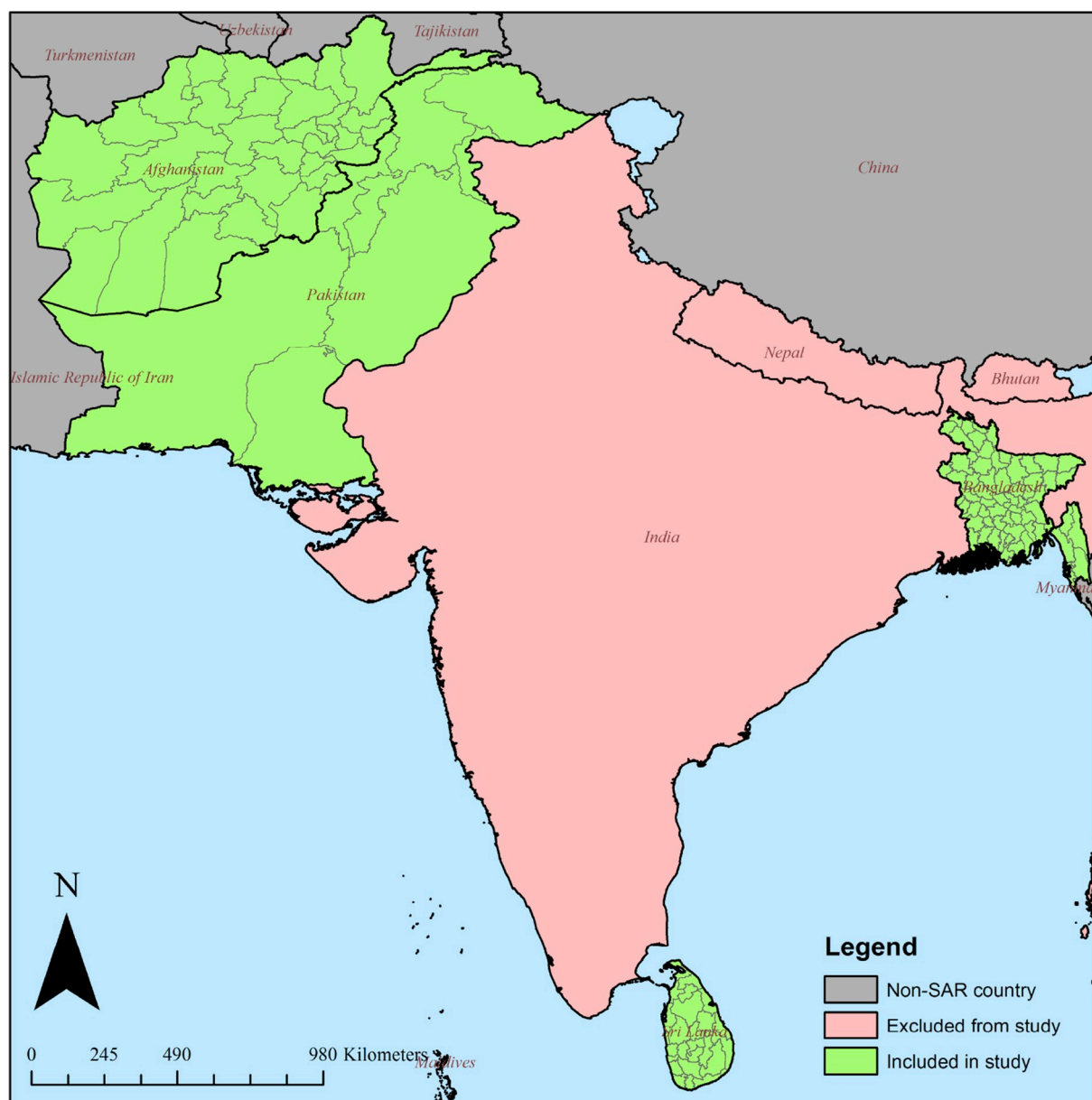


Fig. 1. Data geographical coverage.

all 389 districts across 34 provinces.⁷

Apart from those already discussed above, the CPI and household survey price data differ in other ways, such as differences in the number of food items listed for each food group. This is particularly important, because having fewer food items listed under each food group might mean that it would be less likely that we capture the lowest-cost food items to calculate CoRD, if the few items listed tend to be the costlier food items (see Annex Table A1). Across countries there are also differences. From the household surveys we use the number of food items listed in Sri Lanka was 140, in Pakistan 69, in Afghanistan 61, and in Bangladesh 82.

4. Food-based dietary guidelines (FBDGs)

The second piece of information that is needed to calculate CoRD

are FBDGs. We reviewed FBDGs for five countries in South Asia which had national FBDGs: Afghanistan, Bangladesh, India, Nepal, and Sri Lanka. All dietary guidelines were sourced from the FAO repository of food-based dietary guidelines (FAO 2019). Of the five countries, only Bangladesh, India, and Sri Lanka have quantitative guidelines⁸ (See Fig. 2 for the Sri Lanka food guide).

The guidelines share similarities in how food is categorized. All countries include starchy staples, vegetables, fruits, pulses, dairy foods, other animal-source foods (fish, meat, eggs), and fats and oils in their groupings. In addition, three countries include sugar and sweets as a food group to limit. Nuts and seeds are grouped together with pulses in the dietary guidelines of Afghanistan, while they are grouped together with fats and oils in the guidelines of India and Sri Lanka, and not mentioned in those of Bangladesh or Nepal. Excluding the sweets category, most countries have six food groups: starchy staples, vegetables, fruits, “protein foods” (pulses, fish, meat, eggs), dairy, and fats and

⁷ We had used the 2011 ALCS for Afghanistan, because the 2013 NRVA did not collect food prices, and the data for the most recent household survey (the 2016-17 ALCS) has not yet been released.

⁸ Afghanistan only has key messages and a food guide, but no quantitative amounts or guidelines document yet available.

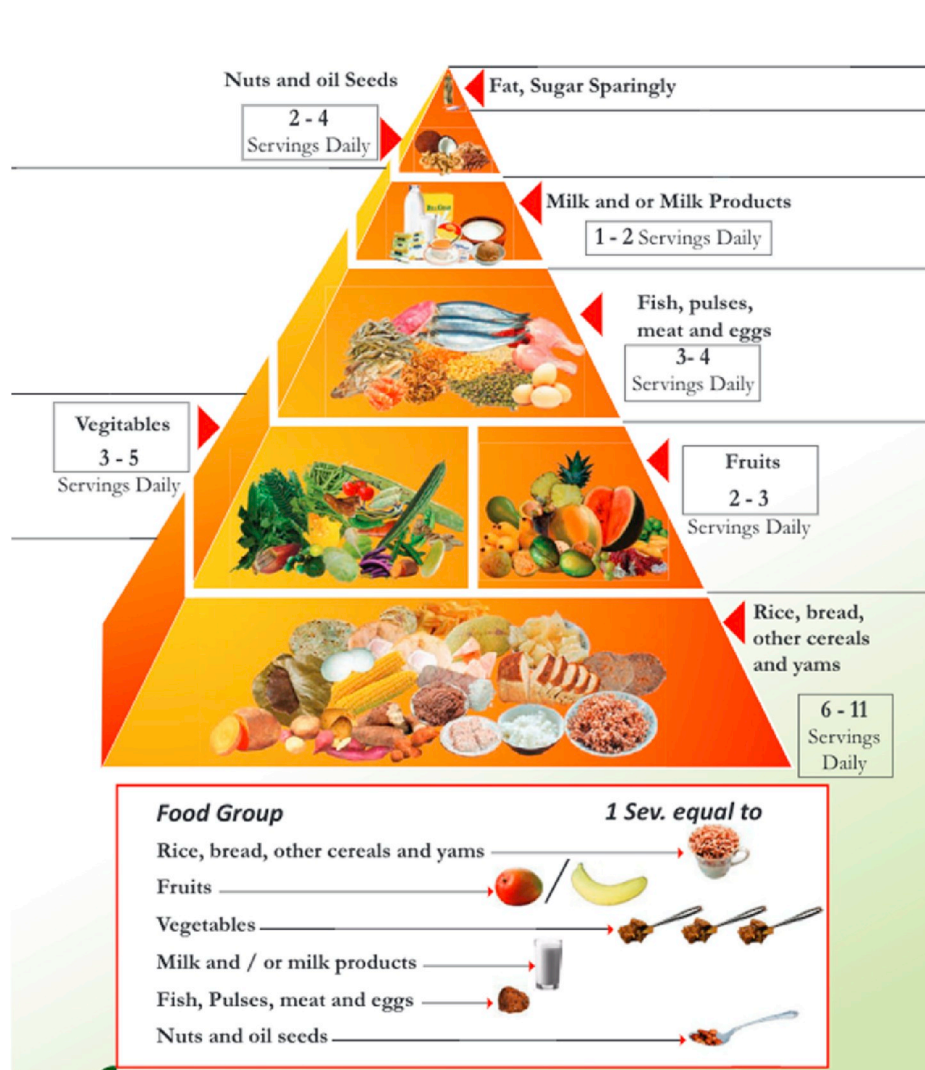


Fig. 2. Sri Lanka food Guide. Source: Food based dietary guidelines for Sri Lankans 2011.

oils.⁹

In attempt to consistently calculate CoRD across these countries, we developed a set of regional FBDG based on similarities among and averages of the national FBDG of the five countries which have some version of the guidelines. To calculate CoRD, we need to assign recommended amounts to each food group. For this, we rely on the quantitative FBDG from Bangladesh, India, and Sri Lanka. Table 2 shows the regional dietary guidelines generated from these countries' national dietary guidelines (Bangladesh Institute of Research and Rehabilitation in Diabetes, Endocrine and Metabolic Disorders (BIRDEM, 2013; Indian Council of Medical Research National Institute of Nutrition, 2011; Sri Lanka Nutrition Division Ministry of Health, 2011)).

5. Results

5.1. Cost of a recommended diet (CoRD)

The estimated CoRD for each country is presented in Table 3. Using

⁹ Afghanistan has six, splitting the protein foods into two categories: “pulses” and “meat/fish/eggs”; Nepal has four, lumping “vegetables and fruit” into one category; and India has three, lumping into categories “vegetables and fruit,” “cereals and pulses,” and “dairy and animal foods.”

the household survey data, we estimate that CoRD was 149 SL Rs 149 in 2016 (or \$2.8 in 2011 \$PPP) per adult per day in Sri Lanka, 71 Prs in 2015-16 (or \$2.0 in 2011 \$PPP) in Pakistan, 44 Af in 2011 (or \$2.5 in 2011 \$PPP), and 54 Taka in 2016 (or \$1.7 in 2011 \$PPP).¹⁰ CoRD is higher in Sri Lanka and Afghanistan, and lower in Pakistan and Bangladesh. For Sri Lanka and Pakistan note that the CPI measure for CoRD is about 20% larger than the household survey measure. The opposite is true for Afghanistan where the CPI measure is smaller than the CoRD measure, this might be due to the fact that the prices from the Afghanistan household survey comes from a market price survey as opposed to deduced unit values. From the household survey data, we further calculate the proportion of households which spend less on food than the estimated CoRD. For the respective years, our estimations suggest

¹⁰ PPPs (Purchasing Power Parities) released by the International Comparison Program (ICP) are a more direct measure of what money can buy than exchange rates when local prices are compared internationally. For comparability, we convert the CoRD which is in LCU into 2011 PPP\$ by deflating the CoRD to 2011 prices and then using the 2011 PPP\$ conversion factor for private consumption. The Afghanistan conversion differs in a few ways. First, the 2011 PPP \$ conversion factor for Afghanistan is an official estimate. Second, since the data from Afghanistan is from 2011 (as opposed to 2016 for all the other countries), we do not deflate Afghanistan's CoRD LCU. The GDP deflator data is from the IMF world economic outlook and the 2011 PPP-conversion data is from the International Comparison Program.

Table 2
National food-based dietary guidelines, and decision on regional food-based dietary guidelines.
Source: Compiled by authors from Dietary Guidelines for Bangladesh 2013; Dietary Guidelines for Indians – A Manual 2011; Food Based Dietary Guidelines for Sri Lankans 2011.

Food Group	Bangladesh		India		Sri Lanka		Average		Decision for Regional FBDG
	Min	Max	Min	Max	Min	Max	Min	Max	
Starchy staples	270g (dry)	450g (dry)	270g (dry)	600g (dry)	300g (dry)	550g (dry)	280g	533g	Use India's guideline amounts (widest range); require 2 unique starches <i>Serving size = 30g</i> <i>Min servings = 9</i> <i>Max servings = 20</i>
Vegetables	300g (at least 100g DGLV)	600g	300g (at least 100g DGLV)	300g (at least 100g DGLV)	3 cups (raw) = 90–390g	5 cups (raw) = 150–650g	300g	433g	Use Bangladesh's guideline amounts (widest range); require 3 unique vegetables, including one DGLV <i>Serving size = 100g</i> <i>Min servings = 3</i> <i>Max servings = 6</i>
Fruits	100g (one citrus, one vit A-rich)	300g (one citrus, one vit A-rich)	100g	100g	1 whole (120g)	2 whole (240g)	107g	213g	Use rounded average, 100–200g; require 2 unique fruits <i>Serving size = 100g</i> <i>Min = 1</i> <i>Max = 2</i>
Pulses/Fish/Meat/Eggs	1/3–1/2 c (60–100g) pulses + 1 piece fish or meat (50g?)	1/2 c pulses (100g) + 4 piece fish or meat (200g?)	60g pulses DG says aim for 100–200g/week fish, but no flesh food required (50g/serving)	120g pulses DG says don't exceed 3 eggs/week	3 servings fish, pulses, meat or eggs (90g total)	4 servings fish, pulses, meat or eggs (120g total)	70g pulses or 50–90g ASF	107g pulses or 120–200g ASF	Use Sri Lanka's guideline amounts (clearest and most comprehensive across protein sources). Require 2 unique protein sources and do not require meat. <i>Serving size = 30g for pulses, 50g for meat/fish/egg</i> <i>Min = 3</i> <i>Max = 4</i>
Dairy	100–150 ml	200 ml	300 ml	300 ml	200 ml	400 ml	200 ml (or 60g milk powder)	300 ml (or 90g milk powder)	Use mean, 200–300 ml; require 1 dairy source (e.g., powdered milk, fresh milk, or yogurt) <i>Serving size = 100 ml (or 30g milk powder)</i> <i>Min = 2</i> <i>Max = 3</i>
Fats and Oils	30g	60g	20g	40g	2 servings nuts or oilseeds (30g total)	4 servings nuts or oilseeds (60g total)	30g	60g	Use mode, 30–60g <i>Serving size = 15g</i> <i>Min = 2</i> <i>Max = 4</i>

Note: Bangladesh and Sri Lanka use the same image for their food pyramids, although recommended servings differ slightly.

Table 3
Cost of a recommended diet and mean household food expenditure.

Country	CPI data 2017		Household survey data				Percent of households spending less on food than CoRD
	CoRD LCU	CoRD 2011 PPP\$	Survey	CoRD LCU	CoRD 2011 PPP\$	Mean per adult male equivalent daily food expenditure	
Sri Lanka	178	3.2	HIES 2016	149	2.8	2.7	64%
Pakistan	86	2.3	HIICS 2015-16	71	2.0	1.4	83%
Afghanistan	44	1.9	ALCS 2011	44	2.5	3.3	41%
Bangladesh	NA	NA	HIES 2016	58	1.7	1.8	53%

Notes: LCU refers to Local Currency Units.

that 64% of the population spent less on food than CoRD in Sri Lanka, 83% in Pakistan, 41% in Afghanistan, and 53% in Bangladesh.^{11 12} In Annex Table A5, for reference, we also present comparisons of the share of CoRD to food expenses vs to total expenses.

Recall that in calculating CoRD, we average the two lowest cost items in each food group. The two lowest-cost items for each country are listed in Annex Table A2, which shows the CoRD-generated food basket. The items selected are for the most part reasonable, and very rarely does the method select a less-nutritious item (i.e. condensed milk for Sri Lanka). While we could create new ad-hoc rules in the method to exclude the selection of such items, doing so would blur the lines between the simplicity that the CoRD method offers and the complexity of say the CoD method.

To ex-post validate the soundness of this CoRD basket in its entirety, we calculate to what extent this basket meets the Estimated Average Requirement (EAR) for energy and various nutrients for a 30-year old male (see Annex Table A3). For Sri Lanka, Pakistan, Afghanistan, and Bangladesh, the CoRD basket meets 99% of the energy (Kcal) requirements, and further meets various nutrient requirements. The only two nutrients that are consistently unmet across countries are vitamin B12 and calcium, presumably because animal-source foods beyond dairy, which are rich in these nutrients, were not selected in the CoRD baskets (because pulses were the cheapest form of protein foods). These nutrient gaps could be filled by adding or substituting a small amount of fish (especially whole small fish), meat, or egg to the CoRD baskets, although doing so would increase costs above the estimates we present here.

The method of choosing the two lowest cost items for each food group allows us to arrive at a *minimum* cost of meeting FBDGs. We also present alternative CoRD results when using the average cost across all items in a food group or the median cost across all items in a food group, which might be a closer representation of what it would cost households to meet guidelines in a way that accounts for taste preferences (see Annex Table A4). Notably, these alternative calculations generate much higher values of CoRD. For example, in Sri Lanka, CoRD would have been 167% higher if we used the average across all food items as opposed to the average for the two lowest-cost food items in each food group.

Additionally, we present alternative calculations of CoRD for the household survey data, we use: 1) average unit values (across households), 2) median unit values, and 3) average and 4) median unit values

but with the food items in (3) and (4) being restricted to be the same (lowest-cost) food items selected when using the prices from the CPI data (See Annex Fig. A1). The latter two calculations do not necessarily select the same food items as the former two calculations, because the food lists from the household surveys are similar to, but not exactly the same as the CPI lists (see Annex Table A2). CoRD calculated using median unit values, our preferred measure, is our lowest estimate of CoRD across methods and datasets. As such, throughout this paper we present our lowest-bound (i.e. most conservative) estimate of CoRD.

5.2. CoRD and actual spending by food group

In each of the four countries, majority of the population currently spends less on food than what it would have cost to meet the recommended dietary guidelines. In this section, we highlight that not only are households underspending on food overall, but also that on average households are overspending on certain food categories, relative to the recommended amounts.

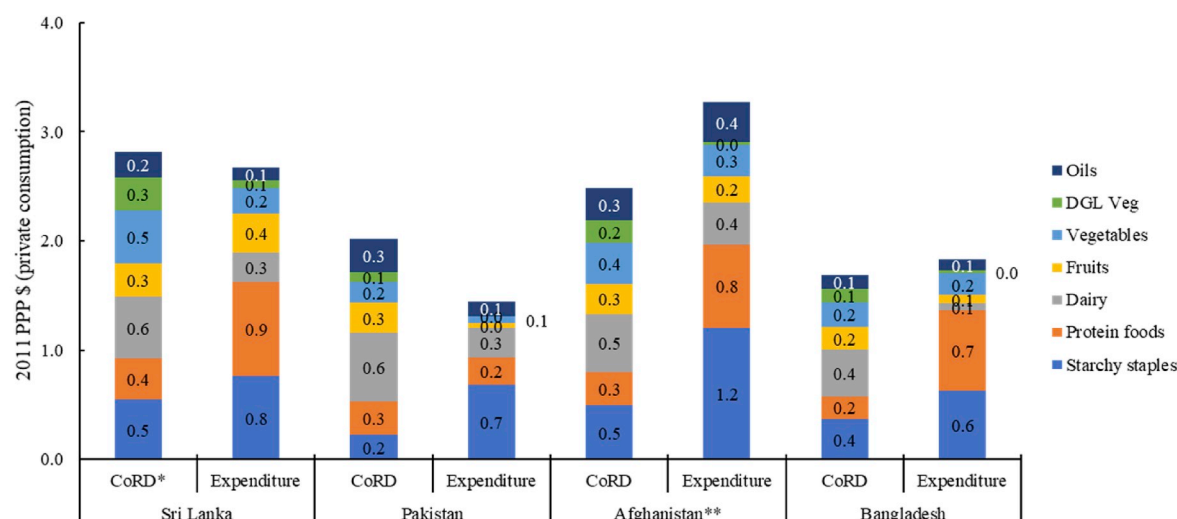
In Fig. 3, for each of the countries we present CoRD and mean household food expenses by each of the food categories. First, we highlight that there is overspending on two food groups relative to minimum cost of recommended amounts: starchy staples and protein foods. Households in Sri Lanka spent \$0.3 more than the CoRD estimate for starchy staples per adult per day, \$0.5 more in Pakistan, \$0.7 more in Afghanistan, and \$0.2 more in Bangladesh. Households in Sri Lanka and Afghanistan spent \$0.5 more than the CoRD estimate for protein foods per adult per day, in Bangladesh \$0.6 more (in contrast in Pakistan, households spend less on protein foods than the CoRD requirement). The overspending on starchy staples and protein foods are, however, slightly different stories.

Overspending on starchy staples may be an adaptation to meet caloric needs, where consumption of other food groups is low. Overspending on protein foods is probably related to consumption of more expensive sources of protein foods, foregoing the option to consume cheaper sources of protein foods. This overspending may well be nutritionally reasonable, if some animal-source foods are selected to meet nutritional needs such as for vitamin B12, which would not be met by always selecting pulses in a strictly least-cost diet. In Fig. 4, we present the FBDG required grams for each food group and the actual per AME consumption in grams. For all four countries, the average household consumes more grams of starchy staples than the FBDG recommended. For protein foods, however, Pakistan and Afghanistan fall short of the recommended grams of protein foods. This implies that an average household in Afghanistan overspends on protein foods without even meeting the recommended protein grams of consumption. Note that CoRD, a calculation of minimum cost, will select cheaper sources of protein foods. For example, instead of meat, the protein foods selected for Afghanistan were mung and lentils (see Annex Table A2).

The overspending on starchy staples and protein foods is mirrored by underspending on dairy and vegetables. Across all countries, relative to CoRD there is underspending on dairy—in Sri Lanka, Pakistan, and Bangladesh by \$0.3, and in Afghanistan by \$0.1. While one might argue

¹¹ For Sri Lanka, Pakistan, and Bangladesh, food expenses of each household calculated as the sum of expenses for each of the CoRD food group categories (that which excludes sweets, spices, and beverages). For Afghanistan, due to data constraints, for each household we instead take the official total food expenses and subtract the proportion spent on the non-CoRD categories sweets, spices, and beverages.

¹² We deliberately avoid making comparisons to the international poverty line of \$1.90 per day (2011 \$PPP). Our calculations are cruder and should be used to provide a broad description of access to a nutritious diet, as opposed to a measure of deprivation.



Notes: ** For Afghanistan, for each household, the expenditure for certain food item was calculated by multiplying quantities consumed with price imputed at district level. The price data was collected from a market price module which is separate from the household food consumption module. We then calculate the share of expenditures on each food group, subsequently applying this composition to "real food expenditure" acquired from harmonized poverty dataset to get "real expenditure" on each food group in LCU.

Fig. 3. CoRD and household food expenses by food group. Notes: ** For Afghanistan, for each household, the expenditure for certain food item was calculated by multiplying quantities consumed with price imputed at district level. The price data was collected from a market price module which is separate from the household food consumption module. We then calculate the share of expenditures on each food group, subsequently applying this composition to "real food expenditure" acquired from harmonized poverty dataset to get "real expenditure" on each food group in LCU.

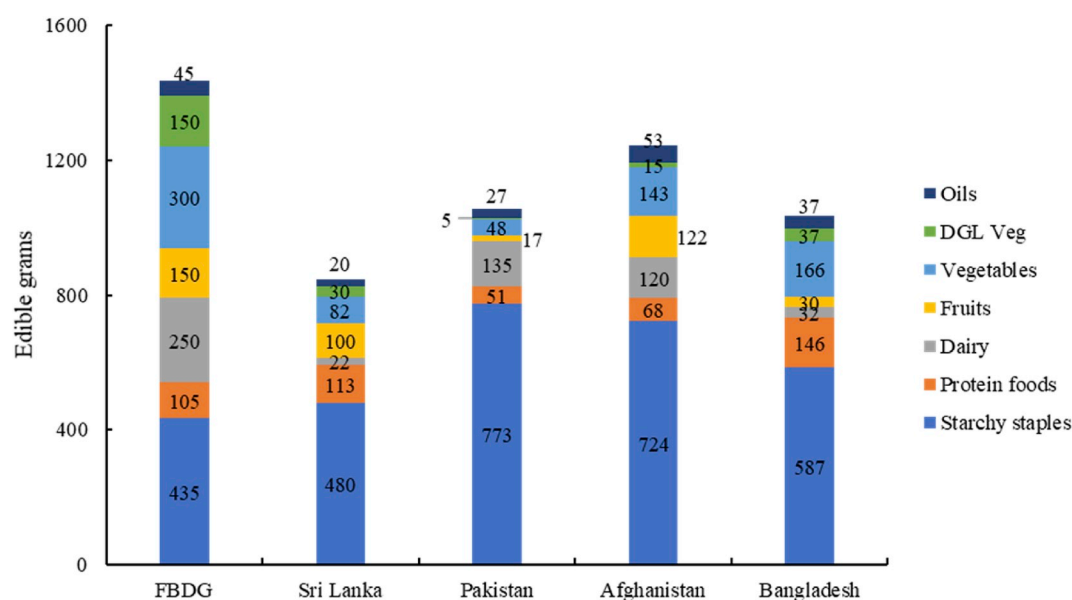


Fig. 4. FBDG and household food consumption in grams by food group.

that households are substituting dairy for protein foods (as both are high in protein), the fact remains that the consumption of both protein and dairy foods fall short of the recommended grams of consumption (see Fig. 4). Households are also largely underspending on vegetables. Looking at vegetables and dark-green leafy vegetables combined, households in Sri Lanka spend \$0.5 less than the CoRD estimate, in Pakistan \$0.2 less, in Afghanistan \$0.3 less, and in Bangladesh \$0.1 less. In other terms, households are spending only 40% of the CoRD estimate of the minimum cost of vegetables in Sri Lanka, in Pakistan only 25%, in Afghanistan only 54%, and in Bangladesh only 65%. Not surprising, therefore, the consumption of vegetables falls much below the recommended grams (see Fig. 4). Households in Sri Lanka consume only 25% of the recommended amount of vegetables, in Pakistan only

12%, in Afghanistan only 35%, and in Bangladesh only 45%.

5.3. CoRD at a subnational level

To map subnational differences in the affordability of CoRD, we simply calculate the proportion and number of households who spend less on food than CoRD— what it would have cost at minimum to meet the recommended diet. For each country, we make this calculation at the lowest subnational level for which the household survey is representative. In Sri Lanka this corresponds to 25 districts, in Pakistan four provinces, in Afghanistan 34 provinces, and in Bangladesh 64 districts.

Figs. 5A, B, C, and D present maps on the estimated proportion of

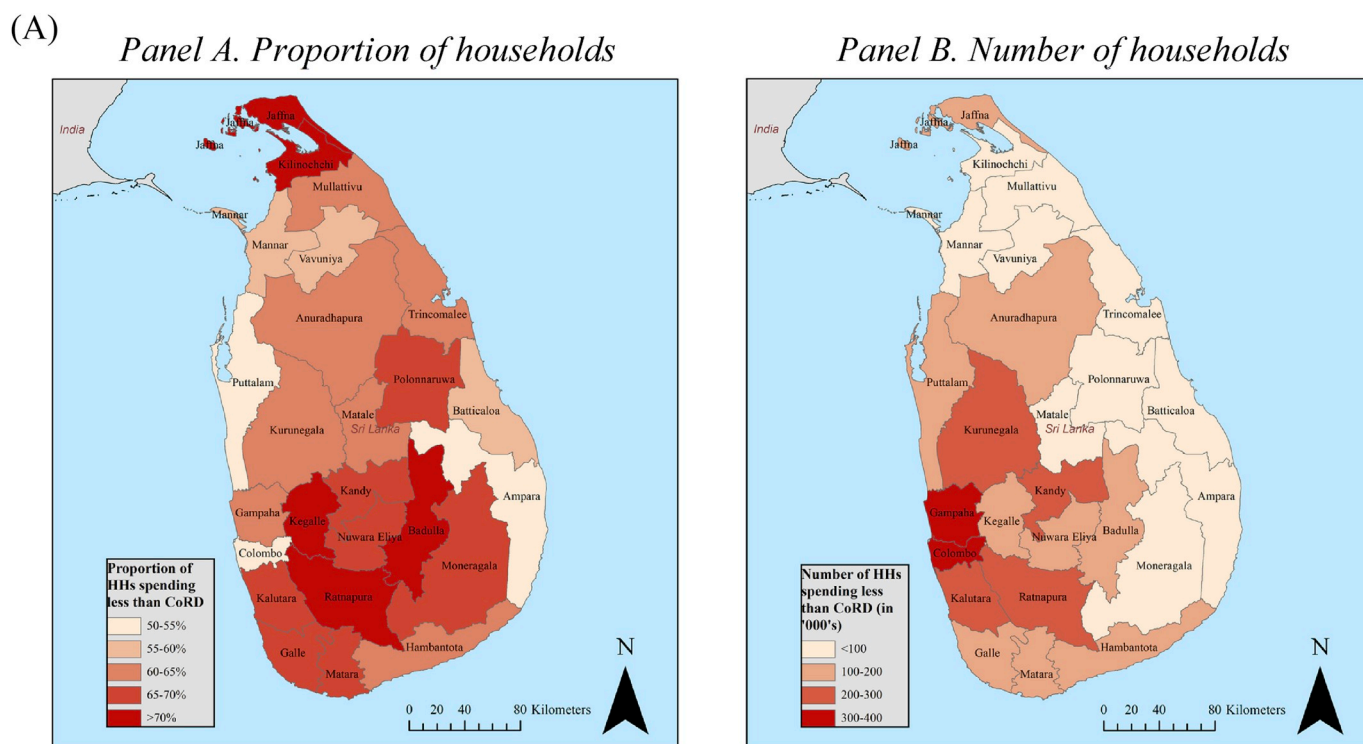


Fig. 5A. Spatial variation in households spending less on food than CoRD in Sri Lanka.

households and number of households who spend less on food than CoRD at the corresponding subnational level for each country. The proportion of households who spend less on CoRD ranges between 53% to 76% across districts in Sri Lanka, 74%–93% across provinces in Pakistan, 6%–83% across provinces in Afghanistan, and 22%–95% across districts in Bangladesh.

Due to population size differences, areas with a higher proportion of low food spending are not necessarily those with a high number of households with low food spending. In Sri Lanka, while Colombo and Gampaha do not have the highest rates of low spending households,

together they constitute over half a million low spending households. This narrative is similar for Punjab in Pakistan, Kabul in Afghanistan, and Dhaka and Chittagong in Bangladesh. Highly populated locations are therefore, diverse, with many households unable to afford the recommended diet, and many others able to.

We next explore which food groups are driving the variability in CoRD across subnational areas for each country (see Fig. 6). For all countries, the coefficient of variation (CV) of the cost of starchy staples, protein foods, and oils is low relative to other food groups (except protein foods in Sri Lanka, and starchy staples in Pakistan for which we have less confidence in the

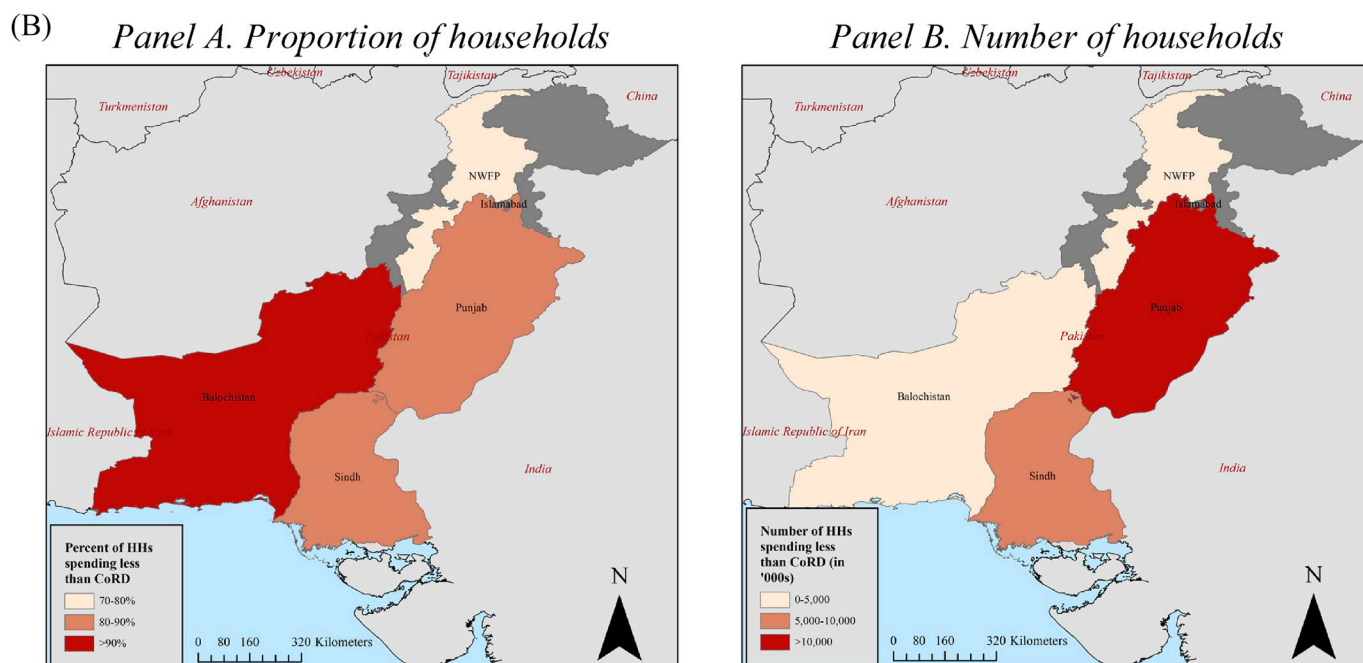


Fig. 5B. Spatial variation in households spending less on food than CoRD in Pakistan.

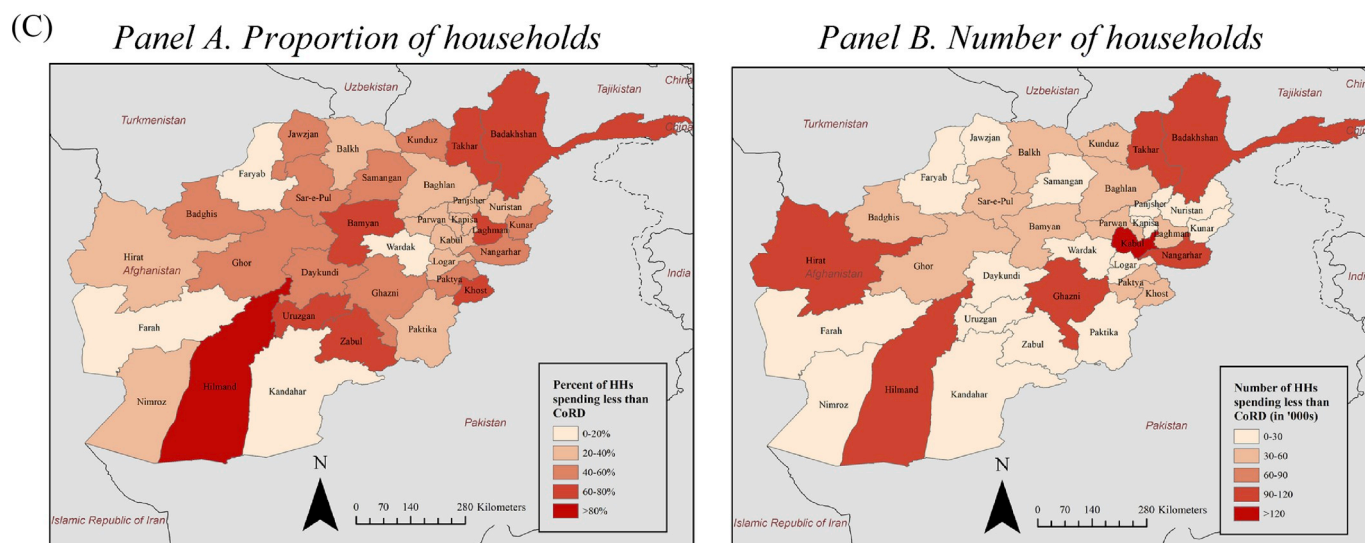


Fig. 5C. Spatial variation in households spending less on food than CoRD in Afghanistan.

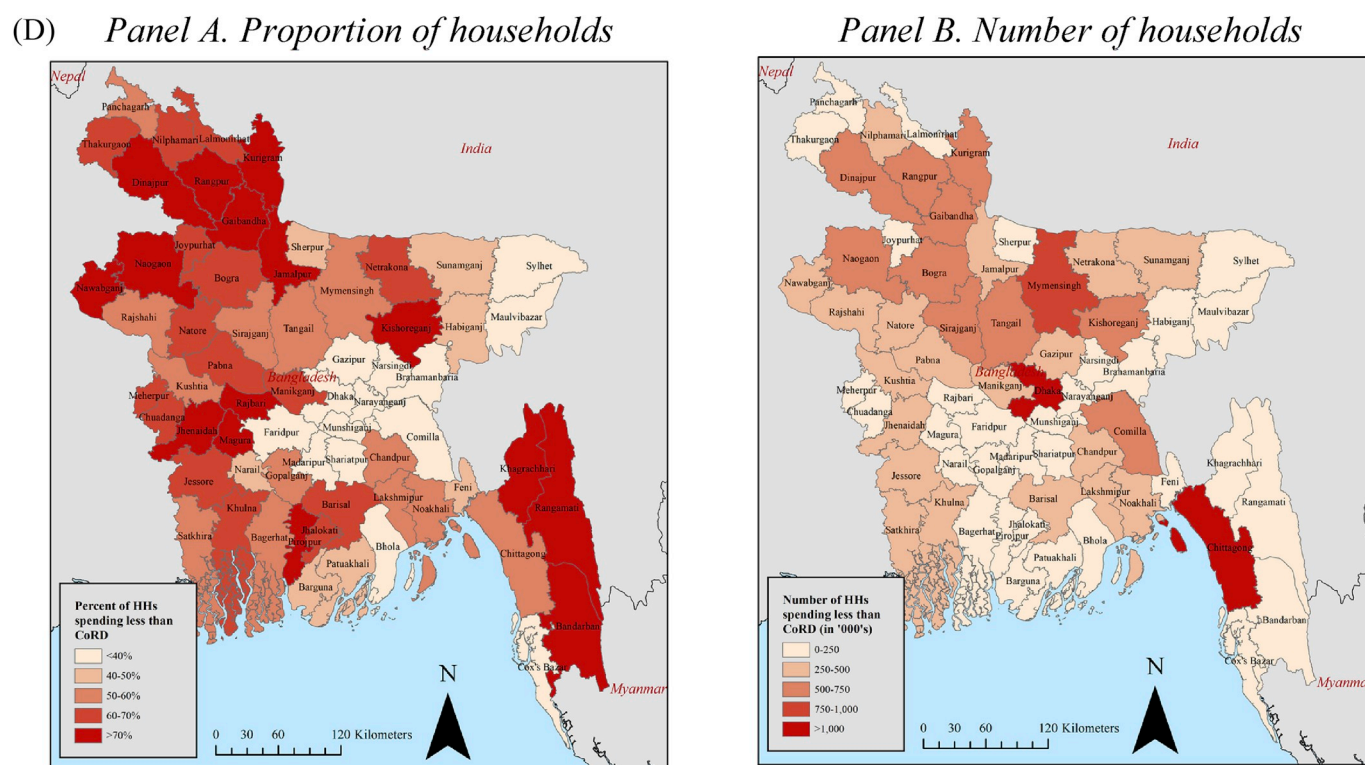


Fig. 5D. Spatial variation in households spending less on food than CoRD in Bangladesh.

estimate given that we are only looking at variation across four provinces).¹³ Starchy staples, protein foods (specifically the non-meat sources of protein foods selected by CoRD), and oils thus tend to have more integrated markets as they are more easily transported within each country. In contrast, the CV of vegetables (and particularly dark-green leafy vegetables) tends to be higher relative to other food groups. Again, this would suggest that vegetable markets are less integrated, and this can be linked to the difficulty in transporting perishable yet nutritious food. Notably, the story is

clearest in Bangladesh, where dairy, fruits, and vegetables have a high CV relative to starchy staples, protein foods, and oils. There are cross-country differences, too. In Sri Lanka and Pakistan, the variation in prices of dairy is much lower, suggesting, for example, that the cold chain might be relatively better under control in these countries, relative to in Afghanistan and Bangladesh. Relatedly, a recent study shows that countries which are more urbanized and have shorter rural travel times to cities face lower retail prices for nutrient-rich foods, similarly suggesting that improving supply chains for perishable foods can lower their prices (Alemu et al., 2018).

6. Discussion

By leveraging data from four countries in South Asia—Sri Lanka, Pakistan, Afghanistan, and Bangladesh—we show that the minimum

¹³ Coefficient of variation (CV) is defined as the ratio of the standard deviation σ to the mean μ . This standardized measure describes the variability relative to the mean of a distribution, which enables cross-country comparison of food price variations under this paper's context.

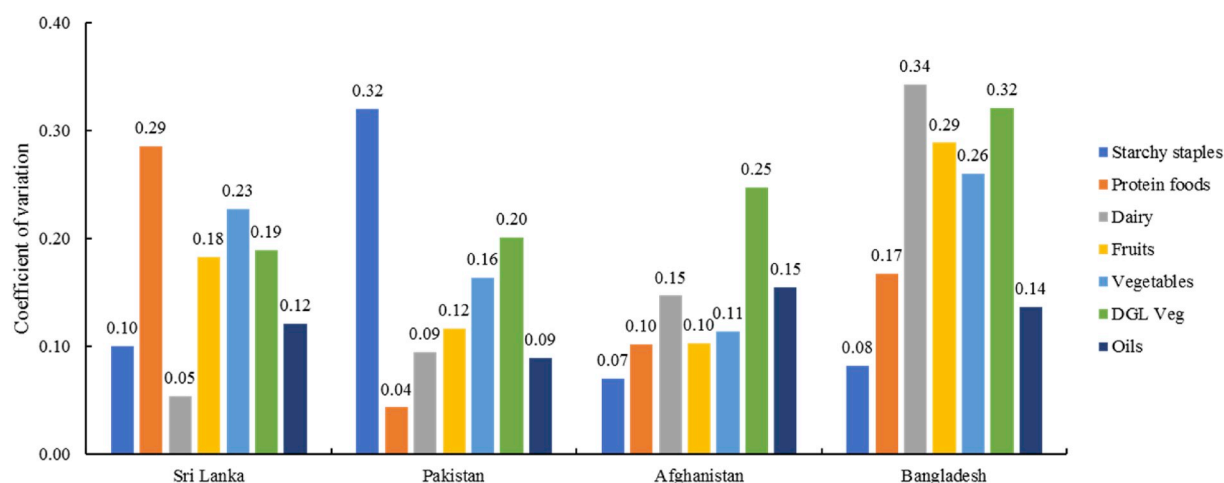


Fig. 6. Coefficient of variation of food group CoRD across subnational level.

cost of meeting national food-based dietary guidelines (CoRD) is high compared to what an average household in these countries currently spends on food. Our estimations suggest that 64% of the population spent less on food than CoRD in Sri Lanka, 83% in Pakistan, 41% in Afghanistan, and 53% in Bangladesh. Moreover, an average household overspends on starchy staples and protein foods (relative to the minimum cost for meeting the recommended diet), and underspends on dairy and vegetables. Spending more than the absolute minimum on protein foods may reflect household purchases of animal source foods, which are more expensive than the plant source protein foods selected in the CoRD basket. This may well be positive for nutrition in order to meet specific nutrient shortfalls (notably vitamin B12; see Table A3) from a least-cost CoRD basket that selects low animal source food (only dairy). As such, on average, households do not meet the recommended consumption for almost all food groups except for starchy staples.

We show that the proportion of households who spend less on food than CoRD varies across subnational areas. These are households who seem to be more constrained by food prices in achieving the recommended diet. While part of this is due to variation in incomes across space, some of this can be attributed to variability in food prices across space. Particularly, the prices of the most energy-dense foods such as starchy staples, (non-meat) protein foods, and oils were the least variable across areas within a country. In contrast, the price of more nutrient-dense vegetables, varied more across areas in the country, suggesting that the market for these vegetables were less integrated. Energy-dense foods generally have a longer shelf life for storage and transportation than nutrient-dense foods, which tend to be highly perishable. It is also important to point out that while the FBDG can technically be met without animal-source protein foods and they are therefore not a focus of the analysis, such foods are also perishable and are likely to have higher price variability than starchy staples and leguminous protein foods.

An agenda to monitor the cost of a nutritious diet

Our paper demonstrates an important proof of concept: national statistics organizations (NSOs) can enrich existing food price monitoring systems by easily integrating in and monitoring the cost of a nutritious diet, provided that they have a sufficient diversity of foods from each food group. Costing food-based dietary guidelines is coherent with the overall national policy promotion of such guidelines. Raw price data from the CPI are rarely publicly released. As such, NSOs themselves may be in the best position to monitor the cost of a nutritious diet, as they already routinely collect and analyze food price data. International organizations could partner with NSOs to conduct cross-country monitoring of these trends. Ministries of food and agriculture, which often have a mandate to assure food security, may also

have a stake in tracking these metrics. While not explored in this paper, these ministries often have their own market information system by which they collect food prices. To inform programs and investments at the country level, there may be interest to track the cost of a nutritious diet and the individual food groups therein.

A food policy shift toward nutrient-rich foods

Motivated by the initial concern of food security, agriculture policies have emphasized energy-dense foods for the last several decades. Current global production data in comparison to dietary guidelines show that agriculture under-produces vegetables, fruits, and protein foods, and over-produces grains, oils, and sugar (Krishna Bahadur et al., 2018). Our findings suggest that food security and agriculture policies need to focus more on nutrient-rich foods, which are relatively unaffordable and lacking in the diet of an average household. In particular, policymakers can reduce the cost and price volatility of a nutritious diet by taking steps to reduce vegetable prices, and by improving the integration of markets particularly for perishable, nutrient-dense foods. However, this is not to say that advocating for a rise in cereal prices is sound. As cereals will tend to be a large share of the budget, an increase in cereal prices tends to negatively consumption of both cereals and non-cereals. Rather, while ensuring stable and low cereal prices has been good policy, ensuring stable and low prices of nutritious foods should be at least of equal priority.

Already, the most nutrient-dense foods are missing from a majority of diets around the world (Micha et al., 2012). This has high costs to human health and productivity. Malnutrition and dietary risks are the top two factors in the global burden of disease; low consumption of fruits and vegetables are high among the top dietary risk factors globally (GBD 2017 Risk Factor Collaborators, 2018). Our results suggest that there may be strong economic reasons for why consumers consume little of the most nutrient-dense foods: They simply cannot afford it. If policymakers seek to improve diet quality and nutrition, then they must focus on improving the affordability of vegetables, fruits, and animal-source foods. This can be achieved likely through a combination of lower prices for nutritious-foods and enhanced incomes for the poorest.

Role of funding source

This paper was funded by contributions from UK Aid and the European Commission, through the South Asia Food and Nutrition Security Initiative (SAFANSI grant number TF0A5366) administered by the World Bank. The funding source had no involvement in the study design; in the collection, analysis and interpretation of data; in the writing of the report; or in the decision to submit the article for publication.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gfs.2019.07.003>.

Annex

Table A1

Number of food items by food group for CPI and HH survey

	<i>Sri Lanka</i>		<i>Pakistan</i>		<i>Afghanistan</i>		<i>Bangladesh</i>	
	CPI	HH	CPI	HH	CPI	HH	CPI	HH
Starchy staples	20	27	7	13	14	12		14
Protein foods	50	48	15	19	21	11		29
Dairy	10	7	4	6	7	6		4
Fruits	14	17	12	11	15	12		16
Vegetables	27	27	17	15	19	13		12
DGL Veg	6	10	1	1	1	3		1
Oils	5	4	3	4	5	4		6
Sum	132	140	59	69	82	61		82

Table A2

Two lowest cost items for each food group

Food group	<i>Sri Lanka</i>		<i>Pakistan</i>		<i>Afghanistan</i>		<i>Bangladesh</i>	
	CPI	HH Survey	CPI	HH Survey	CPI	HH Survey	CPI	HH Survey
Starchy staples	White rice	White rice	Wheat	Wheat	Corn	Barley	Coarse rice	
Protein foods	Red rice	Red rice	Rice	Rice flour	Barley	Maize	Wheat	
	Red lentils	Gram dhal	Red lentil	Pulse gram	Peshawry	Dried peas	Mung	Green gram
Dairy	Soya beans	Watana dhal	Black gram	Gram whole	Vetches	Lentils	Pea gram	
	Fresh cow milk	Cow milk	Curd	Fresh milk	Fresh cow milk	Fresh milk	Liquid milk	
Fruits	Condensed milk	Milk packets	Fresh milk	Yogurt	Imported milk	Yogurt	Curd	
	Mangoes	Mango	Watermelon	Bananas	Watermelon	Mulberries	Guava	
Vegetables	Papaya	Ambarella	Muskmelon	Watermelon	Melon	Mangoes	Amra	
	Plantain flower	Plantain flower	Onion	Radish	Radish	Onion	Pumpkin	
DGL Veg	Squash	Radish	Eggplant	Carrot	Turnip	Carrots	Radish	
	Sweet potato leaves	Gotukola	Spinach	Spinach	Spinach	Spinach	Spinach, Amaranta, Basil	
Oils	Coconut oil	Coconut oil	Cooking oil	Cooking oil	Local butter	Vegetable oil	Soybean oil	
	Vegetable oil	Vegetable oil	Mustard oil	Mustard oil	Imported Vegetable oil	Animal fat	Palm oil	

Table A3

CoRD food basket percent met of Estimated Average Requirement (EAR). for energy and essential nutrients for an adult male

	EAR	<i>Sri Lanka</i>	<i>Pakistan</i>	<i>Afghanistan</i>	<i>Bangladesh</i>
Energ_Kcal	2725 (EER)	99%	99%	99%	99%
Protein_(g)	46.2	151%	174%	193%	189%
Lipid_Tot_(g)	61 (no EAR)	107%	106%	118%	104%
Calcium_(mg)	1000 (AI)	65%	69%	73%	63%
Iron_(mg)	6	249%	337%	417%	272%
Magnesium_(mg)	330	197%	203%	258%	140%
Potassium_(mg)	4700 (AI)	71%	87%	106%	82%
Zinc_(mg)	9.4	125%	143%	179%	104%
Vit_C_(mg)	75	113%	116%	163%	348%
Thiamin_(mg)	1	220%	201%	296%	136%
Niacin_(mg)	12	155%	182%	184%	99%
Vit_B6_(mg)	1.1	241%	278%	281%	145%
Folate_Tot_(µg)	320	236%	219%	319%	347%
Vit_B12_(µg)	2	56%	51%	51%	55%
Vit_A_RAE	625	94%	332%	335%	238%
Vit_E_(mg)	12	22%	125%	69%	99%
Vit_K_(µg)	120 (AI)	397%	624%	643%	657%

From National Nutrient Database for Standard Reference (USDA).

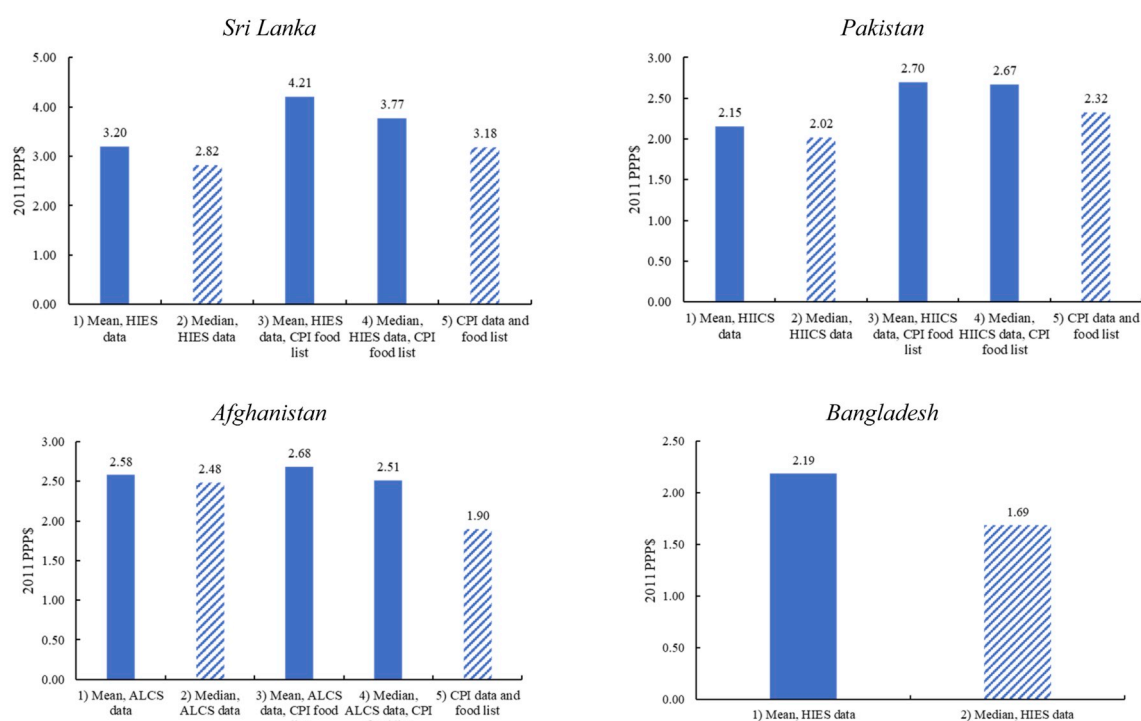
CoRD has 1,435 grams of food.

Table A4
Alternative methodologies of calculating CoRD (LCU)

	Average of two-lowest cost items in each food group (<i>original method</i>)	Average across all food items in each food group	Median across all food items in each food group
Sri Lanka	149	399	332
Pakistan	71	205	127
Afghanistan	44	154	95
Bangladesh	58	117	100

Table A5
Share of CoRD to food expenses vs to total expenses

Country	Survey	Share of CoRD to food expenses	Share of CoRD to total household expenses
Sri Lanka	HIES 2016	106%	23%
Pakistan	HIICS 2015-16	139%	35%
Afghanistan	ALCS 2011	76%	39%
Bangladesh	HIES 2016	92%	35%



Note: Bars with diagonal lines are the values reported in the main text. 1) CoRD results derived using HH survey prices and food list with average food prices being used; 2) CoRD results derived using HH survey prices and food list with median food prices being used; 3) CoRD results derived using HH survey prices but CPI food list with average food prices being used; 4) CoRD results derived using HH survey prices but using the CPI food list with median food prices being used; 5) CoRD results derived using CPI prices and food list. “DGL Veg” = dark green leafy vegetable.

Fig. A1. Alternative specifications of CoRD calculations. Note: Bars with diagonal lines are the values reported in the main text. 1) CoRD results derived using HH survey prices and food list with average food prices being used; 2) CoRD results derived using HH survey prices and food list with median food prices being used; 3) CoRD results derived using HH survey prices but CPI food list with average food prices being used; 4) CoRD results derived using HH survey prices but using the CPI food list with median food prices being used; 5) CoRD results derived using CPI prices and food list. “DGL Veg” = dark green leafy vegetable.

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